Roll No. ___________

Pandit Deendayal Energy University

End Semester Examination – (May, 2024)

B. Tech. (Computer Science and Engineering)

Semester - III

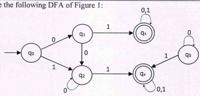
Course Name: Theory of Computation
Course Code: 20CP206T

Date: 08/05/2024
Time: 3 hours
Max. Marks: 100

Instructions:

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Question No.	Description	Marks	Course Outcomes (CO's)
1.	a) Clarify the concept, "Families of Languages" in reference to Theory of Computation with suitable diagram. Provide examples of machines associated with each language family.	7	CO4
	b) Write a context free grammar for a language of even length strings.	6	
	c) Construct a nondeterministic pushdown automaton for a language $L=a^n b^{2n+1} \| n>=1$. **OR** Construct a nondeterministic pushdown automaton for a language $L=0^i 1^j 2^k \| i==j$ or $j==k$; $i, j, k>=1$.	7	
2.	a) Design a Linear Bounded Automaton (LBA) for the language $L = \{ww \mid w \in (0,1)^+\}$ **OR** Design a Linear Bounded Automaton (LBA) for the language $L = \{a^i b^j c^k \mid i*j = k; i, j, k \geq 1\}$	10	CO6
	b) Design a Transducer (TM) to reverse a string consisting of a's and b's.	10	
3.	a) Design a Context Sensitive Grammar for $a^n b^m c^n d^m \mid m, n \geq 1$ **OR** Design a Context Free Grammar for an expression over {[,]} such that (i) every left parenthesis has a matching right parenthesis, (ii) the matched pairs are well nested.	7	CO5
	b) In the given CFG, convert the following production rules into Greibach Normal Form (GNF): $S \rightarrow XB \mid AA$ $A \rightarrow a \mid SA$ $B \rightarrow b$ $X \rightarrow a$	10	
	c) Write an appropriate algorithm to convert a Context Free Grammar (CFG) to Push Down Automata (PDA).	3	

4.	a) Construct a minimal DFA accepting a set of strings over {a, b} in which $\{a^2bwa^2 \mid w \in \{a, b\}^*\}$. Here 'w' is the substring containing alphabets over any number of 'a' and 'b' ranging from zero to infinity.	7	CO1
	b) Let L be the language which is represented by: L = {w ∈ {a, b}*| w contains an equal number of occurrences of ab and ba}. For example, ababa ∈ L (two occurrences of ab, and two of ba), whereas bbaba ∉ L (one occurrence of ab, and two of ba). Provide a regular expression whose language is L.	7	
5.	a) Consider the following context-free grammar to generate arithmetic expressions in one variable a, involving addition and multiplication operations only. The Grammar: **S → a | S + S | S × S** Here, S is the start symbol. Draw all the possible parse trees for the string **a + a × a + a** following the given grammar.	6	CO3
	b) What are the different techniques to reduce a CFG? Use 'Elimination of Unit Production' rule to simplify the CFG consists of following production rule: S → 0A | 1B | C A → 0S | 00 B → 1 | A C → 01	6	
6.	a) Minimize the following DFA of Figure 1:  Figure 1	8	CO2
	b) Provide examples to illustrate the differences between Mealy and Moore machines. Explain how the choice between these models can impact the design and implementation of sequential circuits.	6	

Pandit Deendayal Energy University
School of Technology

End Semester Examination
Programme: B. Tech. (Computer Science and Engineering)
Semester – IV

Date: 09/05/2024
Time: 3 hours
Max. Marks: 100

Course Name: Database Management Systems
Course Code: 20CP208T

Instructions:
1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

Q No.	Description	Marks	COs
1(a)	What is database administration? Explain the roles of database administrator in brief.	5	CO1
1(b)	Explain the distinction between closed and open hashing. Discuss the relative merits of each technique in database applications.	5	CO1
1(c)	Suppose you are given the following requirements for a simple database for the Indian Premier League (IPL): IPL has many teams, each team has a name, a city, a coach, a captain, and a set of players, each player belongs to only one team, each player has a name, position(s) (such as captain, vice-captain, wicket-keeper, batsman, bowler, all-rounder), and a set of injury records, a team captain is also a player, and a game is played between two teams and has a date and a score. Construct an ER diagram for the IPL database. List your assumptions and clearly indicate the cardinality mappings as well as any role indicators in your ER diagram.	10	CO2
2	Consider the following relations: Student (<u>snum</u>, sname, level, age); Class (<u>cname</u>, ctime, room, fid) Enrolled (<u>snum, cname</u>); Faculty (<u>fid</u>, fname, deptid)		CO3
2(a)	Write a SQL query to find the age of the oldest student who has enrolled in a course taught by Prof. Joshi.	4	
2(b)	Write a SQL query to find the names of all juniors (level = JR) who are enrolled in a class taught by Prof. Saurabh.	4	
2(c)	Write a relational algebra query to find the names of students who are enrolled in either "DBMS" or "DAA".	4	
2(d)	Write a relational algebra query to find the names of students not enrolled in any class.	4	
2(e)	Create a table of *Enrolled* in SQL by specifying the primary and foreign key constraint.	4	
3(a)	Consider the universal relation R = {A, B, C, D, E, F, G, H, I, J} and the set of functional dependencies F = {AB -> C, A -> DE, B -> F, F -> GH, D -> IJ}. The attributes are atomic. What are the candidate keys for R? Based on these candidate keys, is this relation in 1NF, 2NF, or 3NF? Why or why not? How would you successively normalize it completely?	10	CO4

<table>
<tr><td>3(b)</td><td>Consider a relation R (A, B, C, D, E, F, G, H) satisfies the following FD's
{A -> B, ABCD -> E, EF -> G, EF -> H, ACDF ->EG}
Find the minimal cover for this set of FD's. Show all the steps clearly.</td><td>5</td><td>CO4</td></tr>
<tr><td>3(c)</td><td>Consider the sample data for a Worker database in a single relation. A worker (workerID, name) has a particular skill (skillType) associated with him. The worker starts working (startDate) in a building (bldgID). Each worker has a bonus rate (bonusRate) associated with him that depends on the skill type of the worker.
<table>
<tr><th>workerID</th><th>name</th><th>skillType</th><th>startDate</th><th>bldgID</th><th>bonusRate</th></tr>
<tr><td>1235</td><td>Henry</td><td>Electric</td><td>10/10/2016</td><td>312</td><td>3.50</td></tr>
<tr><td>1235</td><td>Henry</td><td>Electric</td><td>17/10/2016</td><td>515</td><td>3.50</td></tr>
<tr><td>1412</td><td>Jack</td><td>Plumbing</td><td>01/10/2016</td><td>312</td><td>3.00</td></tr>
<tr><td>1412</td><td>Jack</td><td>Plumbing</td><td>08/12/2016</td><td>460</td><td>3.00</td></tr>
<tr><td>1412</td><td>Jack</td><td>Plumbing</td><td>15/10/2016</td><td>435</td><td>3.00</td></tr>
<tr><td>1412</td><td>Jack</td><td>Plumbing</td><td>10/08/2016</td><td>315</td><td>3.00</td></tr>
<tr><td>1311</td><td>Jane</td><td>Electric</td><td>10/10/2016</td><td>312</td><td>3.50</td></tr>
</table>
Find at least one insertion, deletion and modification anomaly in the above table.</td><td>5</td><td>CO4</td></tr>
<tr><td>4(a)</td><td>During its execution, a transaction passes through several states. List all possible sequences of states through which a transaction may pass with the help of a diagram. Explain why each state transition may occur.</td><td>5</td><td>CO5</td></tr>
<tr><td>4(b)</td><td>For each of the following schedules consisting of three transactions T1, T2, and T3:
a) $S_a = r_1(A); r_2(B); w_1(B); w_2(C); r_3(C); w_3(A);$
b) $S_b = r_1(A); r_2(A); r_1(B); r_2(B); r_3(A); r_4(B); w_1(A); w_2(B);$
Answer the following questions:
i. Draw precedence graph for the schedule by listing all conflict operations.
ii. Is the schedule conflict-serializable? If so, what are all the equivalent serial schedules?
iii. Is the schedule recoverable? Why or why not?</td><td>15</td><td>CO5</td></tr>
<tr><td>5(a)</td><td>Consider a relation R containing one million records, where each block of the relation holds 10 records. R is organized as a heap file (i.e., records are randomly ordered. Assume that attribute a is a primary key for R. For each of the following queries, name the approach that would most likely require the fewest I/Os for processing the query. The approaches to consider are (i) Scanning through the whole heap file for R, (ii) Using a B+ tree index on attribute R.a, and (iii) Using a hash index on attribute R.a. The queries are:
(a.) Find all tuples in R. (b.) Find all tuples in R such that $a < 50$.
(c.) Find all tuples in R such that $a = 50$. (d.) Find all tuples in R such that $a > 50$ and $a < 100$.</td><td>8</td><td>CO6</td></tr>
<tr><td>5(b)</td><td>A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, what will be maximum number of nodes that could be newly created in the process. Justify,</td><td>2</td><td>CO6</td></tr>
<tr><td>5(c)</td><td>Attempt any two from the following questions:
(i) Distinguish between RDBMS and NoSQL. Describe BASE rule.
(ii) Explain the major differences between the Materialization and Pipelining methods of query evaluation.
(iii) Describe the differences between Security and Integrity in databases.</td><td>5+5</td><td>CO6</td></tr>
</table>

Roll No. ___________

PanditDeendayal Energy University

End Semester Examination - May 2024

B. Tech. (SOT-CSE)

Semester - IV

Course Name : Operating System	**Date: 10/05/2024**
Course Code : 20CP207T	**Time: 3 hours**
	Max. Marks: 100

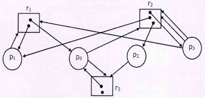
Instructions:

1. Do not write anything other than your roll number on question paper.
2. Assume suitable data wherever essential and mention it clearly.
3. Writing appropriate units, nomenclature, and drawing neat sketches/schematics wherever required is an integral part of the answer.

NOTE: All questions are mandatory to attend however some internal choices are given.

		Marks	CO, Level
Q1.	Define the term semaphore in the context of process synchronization and state its types. What is the role of synchronization primitives such as mutex locks in managing concurrent access to shared resources?	**5 * 2= 10**	**[CO1], L1**
Q2.	Specify Banker's algorithm to avoid deadlock between processes requesting instances of multiple resources. Additionally, specify different strategies to prevent deadlocks in a concurrent computing environment. **OR** State the conditions necessary for the deadlock occurrence among processes. Also explain the mechanism of deadlock ignorance and deadlock avoidance.	**5 * 2 = 10**	**[CO1], L2**
Q3	Explain the concept of Demand Paging in memory management.How virtual memory mechanism facilitates effective memory management within operating systems?	**5 * 2= 10**	**[CO4], L2**
Q4	Consider the set of 6 processes A, B, C, D, E and F, whose arrival time (ms)are 5, 7, 1, 3, 6 and 2, burst time (ms) are 8, 7, 6, 3, 9 and 5. If the CPU scheduling policy is Round Robin with time quantum = 2ms, calculate the average waiting time and average turnaround time. Also state that after what time the process A is completed. **OR** Consider the set of 6 processes A, B, C, D, E and F, whose arrival time (ms) are 0, 5, 12, 2, 9 and 8, burst time (ms)are 11, 28, 2, 10, 16 and 15. If the CPU scheduling policy is priority with preemption, calculate the average waiting time and average turnaround time. Also state that after what time the process A is completed.	**5*1 = 5**	**[CO3], L3**
Q5	Given two concurrent processes i and j, suggest Peterson's algorithm to achieve mutual exclusion between them.Discuss any potential scenarios where Peterson's solution may fail to provide mutual exclusion.	**5 * 2= 10**	**[CO2], L5**
Q6.	What is Process Control Block (PCB)? Explain each parameter of PCB in detail. **OR** Describe how the concept of context switching occurring between processes and threads is different.Justify the claim that thread is lighter in weight than the process.	**10*1 = 10**	**[CO2], L2**

Q7.	(a) What would be the approximate size of the page table required for a system with 64 MB of physical memory and operating within a 32-bit virtual address space, with each page being 4 KB in size? (b) A TLB-access takes 20 ns and a main memory access takes 60 ns. What is the effective memory access time (in ns) if the TLB hit ratio is 80% and there is no page-fault?	**5 *2 = 10**	**[CO4], L3**
Q8.	Given the request queue 95, 180, 34, 119, 11, 123, 62, 64 where the read-write head is initially positioned at the track 50 and the tail track is 199. Consider disk arm movement "towards the smaller values". Draw the scheduling diagrams and calculate total head movement for: (i) Shortest Seek Time First (ii) SCAN (iii) Circular SCAN.	**5 *3 = 15**	**[CO6], L3**
Q9.	Compute the total page faults using First-In-First-Out (FIFO) and Least Recently Used (LRU) algorithms for a reference string: 1 2 3 4 1 2 5 1 2 3 4 5, assuming a system with 4-page frames. Provide a diagram illustrating the page frame allocation at each step for both algorithms.	**5 *2 = 10**	**[CO5], L3**
Q10.	(a) Consider a system with three processes, A, B, and C. They share a total of 4 instances of the same resource type. Each process can request a maximum of k instances. The resource instances can be requested and released only one at a time. Find the largest value of k such that the deadlock is always avoided. (b) Consider the resource allocation graph given in the figure. Find whether the system will result into a deadlock state or not and justify. If there is no deadlock then find a safe sequence.	**5 *2 = 10**	**[CO6], L3**



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Roll No. ___________

Pandit Deendayal Energy University

End Semester Examination: May 2024

B. Tech. (Computer Science and Engineering)

Semester - IV

Course Name: Design and Analysis of Algorithms
Course Code: 20CP209T

Date: 11/05/2024
Time: 3 hours
Max. Marks: 100

Instructions:

1. Do not write anything other than your roll number on question paper.
2. Read question paper properly and assume suitable data wherever essential by mentioning it clearly.
3. Answers written using **PENCIL** will **NOT** be considered for evaluation.

Que. No.	Description	Marks	CO's
Q1	**Select the correct option (1 Mark) and give proper justification (2 Marks) for each of the following Multiple-Choice Questions.**		
Q1 (A)	Two algorithms A and B take $\boldsymbol{n^2}$ **days** and $\boldsymbol{n^3}$ **seconds** respectively to solve an instance of size n. Let β be the smallest instance of n for which algorithm A outperforms algorithm B. Then how much time will be taken by the algorithm A to execute on the size $n = \beta$? i. 2 million years ii. 20 million years iii. 2 billion years iv. 20 billion years	3	CO-1
Q1 (B)	The tight bound to the recurrence $\boldsymbol{T(n) = T(n/6) + T(7n/9) + O(n)}$ is i. $O(n)$ ii. $O(n^2)$ iii. $O(n \log n)$ iv. None of these.	3	CO-2
Q1 (C)	Consider the problem of finding the maximum element of an array using D&C, where we divide the array into two parts, find maximum of each subarray and return the maximum among the two recursively. The number of comparisons required to find the maximum element from the array $[5, 7, 3, 4, 9, 12, 6, 2]$ is i. 4 ii. 5 iii. 6 iv. 7	3	CO-3
Q1 (D)	Which technique is commonly used to prioritize the search paths in Branch and Bound (B&B) for the Travelling Salesman Problem? i. BFS and DFS B&B ii. Greedy strategy iii. Divide and conquer iv. Dynamic programming	3	CO-5
Q1 (E)	You are given two decision problems d_1 and d_2. It is known that d_1 is polynomial time reducible to *clique* and *clique* is polynomial time reducible to d_2. Then which one of the following is more appropriate statement? i. d_1 and d_2 are in NP ii. d_1 and d_2 are in NP Hard iii. d_1 is NP and d_2 is NP Hard iv. d_1 is NP Hard and d_2 is in NP	3	CO-6
Q2	**State the following statements as True or False (1 Mark) and give proper reason to your answer (2 Marks) for each of the following.**		
Q2 (A)	If $f(n)$ is $O(n)$ then $\left(f(n)\right)^2$ is $O(n^2)$.	3	CO-1
Q2 (B)	In the potential method for amortized analysis, the potential energy should never go negative.	3	CO-2
Q2 (C)	Consider the problem of finding the k^{th} smallest number in the array. If we divide an array into groups of 3 (instead of 5), find the median of each group, then recursively find the median of those medians, partition the array according to this median of median and recurse, then we can obtain a linear-time median-finding algorithm.	3	CO-3
Q2 (D)	In a graph with unique edge weights, the spanning tree of second lowest weight is unique.	3	CO-3
Q2 (E)	$3-SAT$ cannot be solved in polynomial time, even if $P = NP$.	3	CO-4

<table>
<tr><td>Q3 (A)</td><td>Define mathematically the below given asymptotic notations with example. Also, discuss why evaluating an upper bound, lower bound and tighter bound is valuable in algorithm analysis.
a) Big O b) Big Omega c) Big Theta</td><td>5</td><td>CO-1</td></tr>
<tr><td>Q3 (B)</td><td>Obtain the guess for the following given recurrence using recurrence tree method.
$T(n) = T(2n/5) + T(3n/5) + n^2$</td><td>5</td><td>CO-2</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td>Solve any two of the following:</td><td></td><td></td></tr>
<tr><td>Q4 (A)</td><td>Solve the following scheduling problem by giving step-by-step computations using greedy strategy. Also, print the job sequence and maximum profit.
<table><tr><td>$deadline$</td><td>7</td><td>2</td><td>5</td><td>3</td><td>4</td><td>5</td><td>2</td><td>7</td><td>3</td></tr><tr><td>$profit$</td><td>15</td><td>20</td><td>30</td><td>18</td><td>18</td><td>10</td><td>23</td><td>16</td><td>25</td></tr></table></td><td>10</td><td>CO-3</td></tr>
<tr><td>Q4 (B)</td><td>a) Write the D & C algorithm for performing large integer multiplication using Karatsuba's trick.
b) Show the multiplication of 3415 and 926 using the above algorithm.</td><td>10</td><td>CO-3</td></tr>
<tr><td>Q4 (C)</td><td>Given a set of coins S and a value V, find the number of ways by which we can make change of V. Here, $S = \{1, 2, 3\}$ and $V = 6$. State the recurrence relation for this problem and show the computation of each value clearly.</td><td>10</td><td>CO-3</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q5 (A)</td><td>Explain how to reduce a 3-SAT to a clique problem. Outline the steps involved in this reduction and describe how to create a graph for the clique problem from a given 3-SAT instance. You can take this instance for your reference:
$F = (x_1 + \sim x_2 + x_3)(\sim x_1 + x_2 + x_3)(\sim x_1 + \sim x_2 + \sim x_3)$</td><td>10</td><td>CO-4</td></tr>
<tr><td>Q5 (B)</td><td>Solve the below given 0/1 Knapsack using Backtracking method. List all the feasible solutions and provide the final optimal solution with corresponding maximized profit.
<table><tr><td>$weight$</td><td>2</td><td>3</td><td>5</td><td>2</td><td rowspan="2">$W = 7$</td></tr><tr><td>$profit$</td><td>10</td><td>20</td><td>30</td><td>40</td></tr></table>
OR
Solve the travelling salesman problem consisting of 4 cities using the least cost branch and bound technique for the given distance matrix - $\begin{bmatrix} 0 & 10 & 9 & 7 \\ 10 & 0 & 5 & 6 \\ 9 & 5 & 0 & 6 \\ 7 & 6 & 6 & 0 \end{bmatrix}$</td><td>10</td><td>CO-5</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q6 (A)</td><td>In the art gallery, we are given a line L that represents a straight hallway in an art gallery. We are also given a set $X = \{x_1, x_2, \dots, x_n\}$ of real numbers that represent locations where paintings are hung in the hallway. Suppose that a single guard can protect all the paintings within distance of at most 1 unit of his/ her position (on both sides). Design a greedy algorithm for finding the placement of the guards that uses minimum number of guards to guard all the paintings with positions in X.</td><td>10</td><td>CO-6</td></tr>
<tr><td>Q6 (B)</td><td>You are given n coins. They all look identical. They should all be the same weight, too but one among them is a fake, i.e. made of a lighter metal. You have a old fashioned balance scale that enables you to compare any two sets of coins. However, for each weighing, you will be charged.
Design a divide and conquer algorithm to find the fake coin in the fewest number of weighing. That is, how many times you must use the scale?
[Hint: This can be easily solved by dividing the coins into 2 sets and it works well. However, you can do better than this. Design the algorithm by dividing the coins into 3 sets.]
Furthermore, how many weighing is required to identify the fake coin from the set of 26 given coins using the above approach?</td><td>10</td><td>CO-6</td></tr>
</table>

========================= **End of the Question Paper** =========================

========================= **Best of Luck** =========================